

Machine learning & Neural Network methods for water quality prediction

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Abstract— Access to safe drinkable water is essential for human survival. However, the increasing pollution of water sources globally has become a major concern, leading to the spread of diseases among populations. Therefore, it is crucial to continuously manage and evaluate the quality of drinking water. While laboratory testing and analysis are currently used to assess water quality, it is a tedious and time-consuming process. Advanced technologies, such as big data and machine learning, have the potential to provide more accurate and efficient results. Several studies have explored the use of such technologies to predict water quality and identify the factors that affect it. For our study, deep learning algorithm was developed using some features, including hardness, pH, chloramines presence, solids presence, sulfate presence, organic carbon, conductivity, turbidity, potability, and trihalomethanes to predict the potability of river water. The model achieved a perfect accuracy of 96% for 20% testing data and 80% training data in predicting water potability.

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I. INTRODUCTION

The significance of freshwater as a crucial natural resource cannot be overstated, as it is essential for the survival of all living organisms. Not only is water necessary for human consumption, but it also plays a vital role in various aspects of daily life, including bathing, cooking, and irrigating crops. As urbanization, economic development, and population growth continue to spread, the demand for water has increased [1]. Despite its importance, ensuring the safety and quality of water is a complex issue. Overuse, climate change, and pollution are among the factors that contribute to the deterioration of water quality. Poor water quality can lead to various water-borne illnesses, emphasizing the need for safe and clean drinking water [2]. Water potability, which refers to water's safety for human consumption, is determined by factors such as the absence of harmful microorganisms and chemicals. Other factors, such as pH levels, clarity, and nutrient abundance, also indicate water quality. Water quality indicators can be categorized into biological, chemical, physical, and radiological factors [3]. Although laboratory analysis has traditionally been the most common method for assessing water quality, advancements in technology have led

to the emergence of more efficient approaches, such as Neural network approach and machine learning [4].

Neural network is a deep learning technique that leverages multiple decision trees to improve accuracy. Neural networks use multiple decision trees, also known as "neurons," to improve accuracy. These neurons are interconnected and work together to process data and make predictions [5]. Neural networks also use randomly selected samples and attributes to create training sets, which are used to train the model [6]. The model then uses these training sets to learn and improve its ability to make accurate predictions. Additionally, neural networks may use techniques such as dropout regularization, early stopping, and weight initialization to further improve accuracy and prevent overfitting [7].

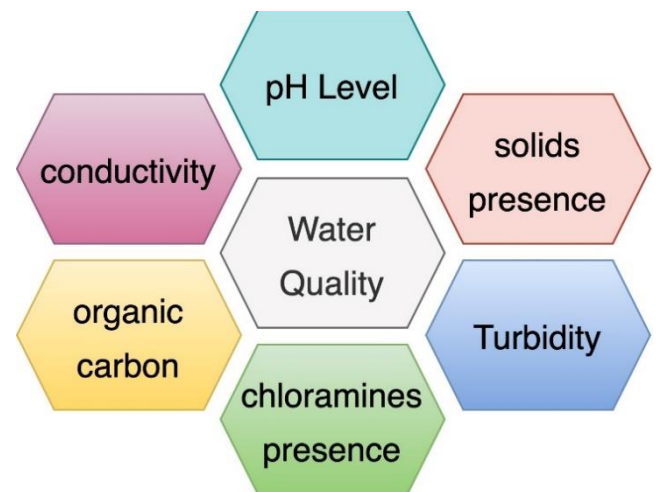


Figure 1 : Water Quality Parameters

This study aims to the factors influence water potability and whether a Neural network model can effectively predict water quality based on these factors. Given the importance of water quality assessment and the potential of Neural network in evaluating water quality, this research is critical.

II. SIMILAR STUDY

Haghiabi et al. [8] developed a model for predicting the quality of water from Tیره river in Iran based on machine learning. The regional water authority collects monthly data to monitor the water quality and stage discharge relation, and these data were used in this study. The chosen parameters for prediction are the specific conductivity, pH, bicarbonate, Sulfer, temperature, Sodium, Calcium, and Magnesium among the dissolved salts. As for the chosen algorithms, these were artificial neural networks ANN where different transfer functions were tested, Support Vector Machine (SVM) where different kernel options were taken into consideration, and group method of data handling GMDH. This method is a feed forward network where each pair of inputs are introduced into one neuron, and the neurons that produce the least error is considered for the next layer development. 80% of the dataset was used for training, and 20% for testing, and the RMSE index was used for evaluation. Each parameter was predicted using the other parameters and the results showed that SVM performs better in predicting all of the parameters, although in occasions all of the three models perform the same. The authors discovered that tansig function for ANN performs well, and that SVM needs the RBF kernel. Finally, DDR index was used and it showed that the three algorithms have a slight overestimation, whereas SVM shows the least data dispersion.

Khoi [9] and his colleagues studied and assessed the performance of a total of 12 machine learning models based on their capability of predicting the quality of water quality. The researchers gathered samples from four different stations around the La Buong River from the year 2010 to the year 2017. The samples were assessed for the following 10 parameters: pH, COD, BOD, temperature, Turbidity, DO, coliform, phosphate, total suspended solid, and ammonium. Based on these measurements, the quality of water can be assessed. The dataset created by these samples was divided into 70% to 30% for training and testing respectively. Five different types of gradient boosting algorithms were chosen for assessment, as well as three decision tree-based algorithms, and four ANN-based algorithms. For feature selection, the model chose the parameters with the highest R2 value, and the selected features were fitted to each algorithm based on trial and error approach. Thus, the overall quantity of model scenarios was 120 models, where 10 different scenarios reflected the inclusion of 1 feature only, two features, until reaching all the features in scenario #10. These models were evaluated based on R2 value and the RMSE value (root mean squared error). The testing showed that XGBoost-S9, Neural network RF-S5, and MLP-S4 scored the highest performance metrics (where S refers to the number of scenario).

Michael E. Omeka [10] aimed to assist farmers in improving their irrigation water management and agricultural productivity. To achieve this, Michael collected data on irrigation water quality and developed a prediction system for the irrigation water quality index. The data was thoroughly cleaned and modified to ensure accuracy and precision. Two methods, Multiple Linear Regression and Artificial Neural Network Modeling, were employed to analyze the features and determine the best model with the highest accuracy. The results showed that both models were suitable for prediction, but the Artificial Neural Network Modeling had the best performance, with R2 values ranging from 0.513 to 0.983. Overall, Michael's research provides valuable insights into improving irrigation water management and enhancing agricultural productivity.

TABLE I. SUMMARY OF SIMILAR STUDIES

Paper	Experimental Environment	Results
El Bilali et al.	The study utilized various machine learning algorithms, including Multiple Linear Regression, ANN, DT, RF, K-KNN, SVM/R, SGD, and Adaptive Boosting.	After conducting tests, it was found that out of the 8 models used, only SVR and K-NN were incapable of predicting all 8 IWQ parameters. The results showed that the adaptive boosting model performed the best among all the models.
Khoi et al.	Five different types of gradient boosting algorithms were chosen for assessment, as well as three decision tree-based algorithms, and four ANN-based algorithms	The testing showed that XGBoost-S9, Random Forest RF-S5, and MLP-S4 scored the highest performance metrics (where S refers to the number of scenario).
Michael E. Omeka et al.	The study employed two modeling techniques, namely Multiple Linear Regression and ANN Modeling.	The best performance was for Artificial Neural Network Modeling with R2 values ranging between 0.513 and 0.983.

III. METHOD

Lately, deep learning has shown impressive effectiveness across various fields such as voice identification, categorizing images and detecting objects, and understanding human language. Researchers have conducted several experiments to enhance the performance of fire detection systems using deep learning, which play a critical role in rapidly and precisely detecting fires and issuing early warning alerts. Swift notifications are essential for preventing damage and saving lives. A variety of detection systems have been evaluated, spanning from conventional mechanical systems to cutting-edge devices that leverage recent technological advancements like computer vision, deep learning, and machine learning [11].

The deep learning approach presents several distinctions compared to the conventional method of fire detection based on computer vision. Firstly, after being trained on a vast amount of diverse data, the network automatically records the features instead of requiring manual investigation by an expert. Consequently, the emphasis has shifted towards developing a suitable network and preparing the training data, rather than attempting to identify the accurate manually crafted features.

Furthermore, the generation of the detector/classifier can be achieved by simultaneously training the neural network's features and detectors. Consequently, the selection of an appropriate network structure becomes even more crucial with a well-designed training approach. DL, ANN, YOLO, CNN, and Transfer Learning (TL) have emerged as the most prevalent computer-based techniques for fire detection.

1) Deep Learning Techniques

As illustrated in Figure 2, DL methods can be categorized into three primary groups:

1. Deep networks utilized for unsupervised or generative learning.
2. Deep networks employed for supervised or discriminative learning.
3. Deep networks utilized for hybrid learning, that combines both of these approaches along with other related techniques.

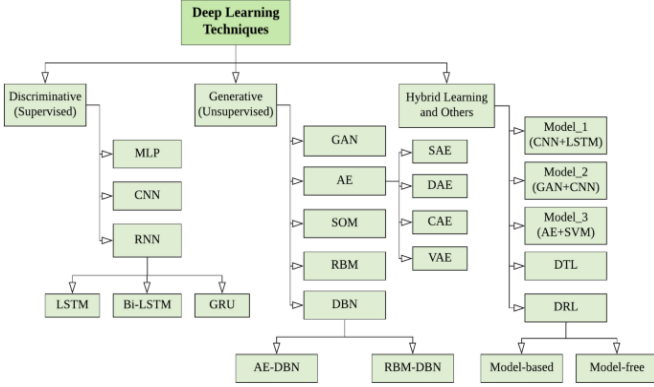


Figure 2 : Deep Learning Techniques

2) Artificial Neural Network

An ANN is a computational model inspired by the operations of actual neurons in a biological nervous system. These simulations closely mimic the behavior of neurons and their ability to transmit electrical signals between the input and output processing of the brain. In essence, ANNs strive to emulate the functionality of biological neurons by employing components that mimic their structure, thereby aiming to create computer systems that operate in a similar manner [12].

The primary goal of an artificial neural network is to generate an output pattern by processing an input pattern. A standard artificial neural network comprises an input unit responsible for storing the inputs, a hidden unit that transforms the inputs into an internal numerical representation, and an output unit that converts the hidden values into predictions. Essentially, an artificial neural network functions as a parallel computing system, composed of fundamental processing components including units, connections, and weights. These elements are interconnected in a specific manner and operate with numerical inputs and outputs to carry out a specific task. Figure 3 depicts a fundamental architecture of an artificial neural network that features multiple inputs and a single output.

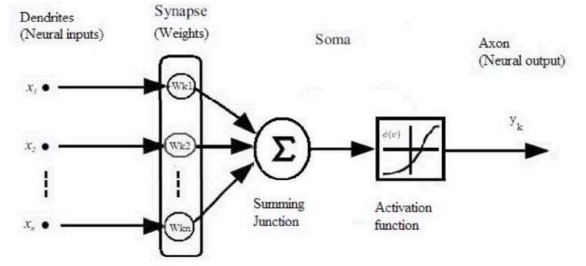


Figure 3 : A simple model of artificial neural network.

3) Activation Function

• Sigmoid Function

The sigmoid function is applicable in cases where values fall within the range of 0 to 1, with 0 representing the lowest probability of an event and 1 representing the highest. This makes it well-suited for predicting events. Furthermore, the sigmoid function's derivative can be calculated as a curve between two points.

$$S(x) = \frac{1}{1 + e^{-x}} \quad (1)$$

• SoftMax Function

The SoftMax function shares similarities with the sigmoid function, but its output values are normalized and can be added up to a total of 1, creating a probability distribution of the output values.

$$\sigma(\vec{z})_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}} \quad (2)$$

• RELU Function

The sigmoid function is being gradually replaced by a new method, where any output value below zero is rounded up to zero, and input values greater than zero are considered equivalent to their respective output values.

$$RELU(x) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases} \quad (3)$$

4) SVM :

Support Vector Machines (SVM) have gained popularity as a supervised machine learning approach for classification and regression tasks. It originated from the pioneering work of Vapnik and Chervonenkis, and its foundation lies in Statistical Learning Theory. One of the remarkable capabilities of SVM is its ability to discern patterns from input data containing both continuous and categorical variables. By constructing a hyperplane in a multi-dimensional space, SVM aims to separate different classes and classify datasets into distinct categories, utilizing the concept of the maximum marginal hyperplane (MMH) [13].

To process input data effectively, SVM employs kernel functions, which facilitate the conversion of data into the desired format. When confronted with non-linear problems, SVM employs the kernel trick technique, incorporating slack variables and additional dimensions to transform the data into a higher-dimensional space. SVM supports a range of kernel types, which are enumerated in Table II.

TABLE II. DESCRIPTION OF WATER QUALITY PARAMETERS

Kernel Type	Equation
Linear	$K(x, y) = (ax^T y + c)$
Sigmoid	$K(x, y) = \tanh(ax^T y + c)$
Radial Basis Function (RBF)	$K(x, y) = \exp\left(-\frac{\ x - y\ ^2}{2\sigma^2}\right)$

A. Proposed System

The figure below shows our system including the steps of method.

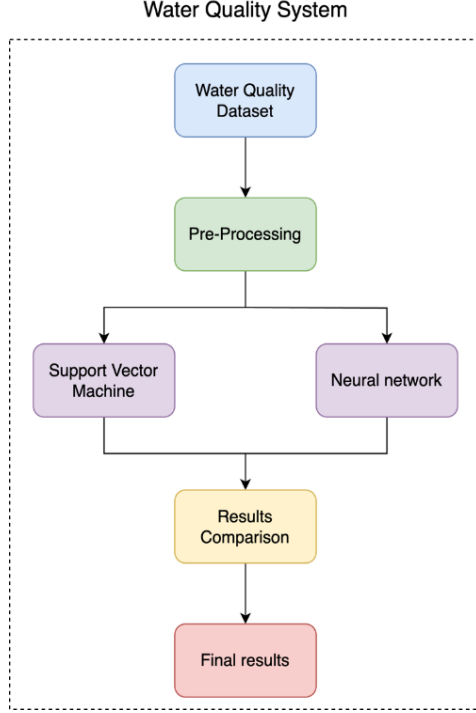


Figure 4 Our proposed method

In this research, deep learning techniques and SVM (Machine learning technique) are chosen to be implemented. The model will combine two separate methods to analyze the quality of water.

Initially, the dataset is selected for water quality. After that, data pre-processing takes place in order to make the input data suitable for the chosen deep learning architectures (neural networks). Finally, the model is evaluated according to accuracy and other metrics, and compared to other methods such as SVM.

B. Dataset Description and Preprocessing

1) Water Quality Dataset

The dataset comprises metrics for 3276 distinct water bodies, each distinguished by its unique water quality. This dataset is commonly utilized to ascertain the potability of the tested water. Table III illustrates the parameters employed in the water assessment process :

Parameter	Description
pH value	Inside water, the acid-base balance is crucial to the well-being of both human and natural life, as any significant deviation from the optimal range can pose a significant threat.
Hardness	The measurement of the concentration of calcium and magnesium salts in water.
Total dissolved solids	The representation of the assemblage of organic and inorganic minerals that are capable of being dissolved in water.
Chloramines	The measurement of residual disinfectants presents in water after the treatment process.
Sulfate	This particular mineral has the potential to occur in groundwater, with its concentration varying based on the geographical location.
Conductivity	The representation of the water's ability to conduct electricity, which is determined by its mineral content.
Organic carbon	The measurement of organic matter content in water, originating from the decomposition of natural or synthetic substances.
Trihalomethanes	The representation of the chemicals that can be identified in water subsequent to chlorine treatment.
Turbidity	Characterizing the light emission properties of water, which are influenced by the concentration of suspended solids.
Potability	Characterizing the suitability of water for human consumption (potability).

2) Data Preprocessing

Data obtained is not always pristine; it often contains null values, missing values, and noise. Such details render the dataset unsuitable for use in Machine Learning or Deep Learning algorithms. Therefore, data preprocessing is performed to clean the data and make it appropriate for inputting into the algorithms.

To ensure the algorithm operates smoothly without any missing data, the first step is to check for the presence of NULL values. These values indicate missing data and can hinder the algorithm's performance. To obtain more accurate results, it's preferable to replace null values. In the provided image, the variables pH, Trihalomethanes, and Sulfate have missing values (NULL). Usually, these null values are substituted with the average or mean of the corresponding category. Following this procedure, the dataset is reexamined to ensure that all missing values have been successfully replaced.

IV. RESULTS

The results related to our proposed work are being presented including the data visualization, as well as the actual results that have been obtained upon implementing our model, according to the selected evaluation metrics.

TABLE III. WATER QUALITY PARAMETER DESCRIPTIONS

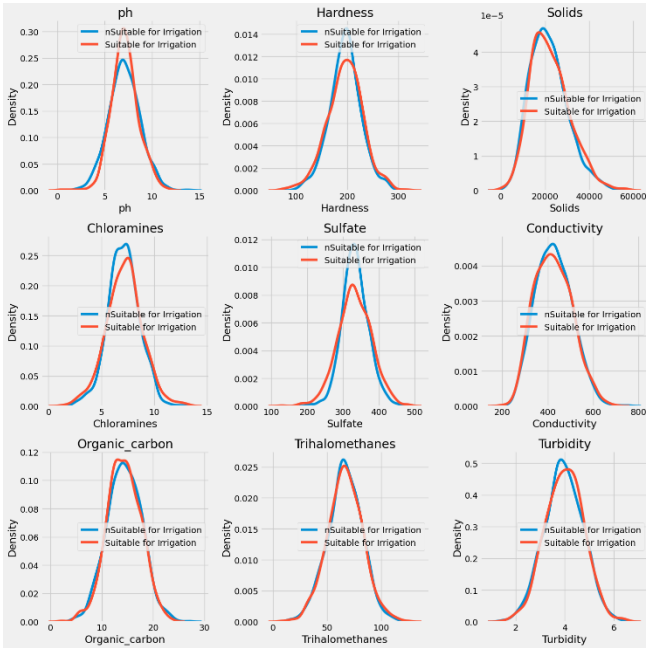


Figure 5 Distribution Parameter for Water Quality

The difference in the proportion of appropriate and inappropriate parameters for each feature is leveraged to forecast the quality of the water. For instance, the classification of tested water as suitable or unsuitable for irrigation is determined based on the contrast between turbidity values, indicating the appropriateness of this parameter.

Precision, Recall, F1-score, and Accuracy are examples of the several metrics that can be used to assess the effectiveness of the models (see references (4, 5, 6, and 7)). Recall, which similar to sensitivity, calculates the true positive rate by measuring the proportion of correctly identified positive inputs out of all inputs that should have been identified as positive. Precision, on the other hand, computes the ratio of accurately classified positive instances to the total positive results. The F-measure combines precision and recall into a single metric, providing a comprehensive evaluation of the model's predictive performance.

Equations:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \quad (4)$$

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \quad (5)$$

$$\text{F-Measure} = 2 * \text{Precision} * \text{Recall} / (\text{Precision} + \text{Recall}) \quad (6)$$

$$\text{Accuracy} = (\text{TN} + \text{TP}) / (\text{TN} + \text{TP} + \text{FN} + \text{FP}) \quad (7)$$

Another significant metric is the area under the curve (AUC), which varies between 0 and 1. A higher AUC value signifies better performance. When the model effectively discriminates between instances of two classes, the AUC value reaches 1. Conversely, if the model struggles to differentiate between any instances, the AUC value diminishes to 0.

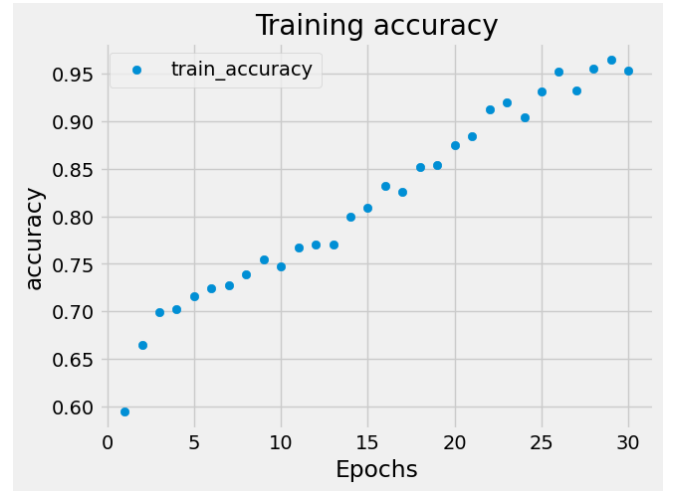


Figure 6 Training accuracy

During the training phase, the proposed water quality recommendation for irrigation system was assessed using evaluation metrics. Accuracy and loss were the criteria employed to evaluate the subsystem. The model exhibiting the lowest loss was identified as the best model and utilized for subsequent evaluation and testing purposes.

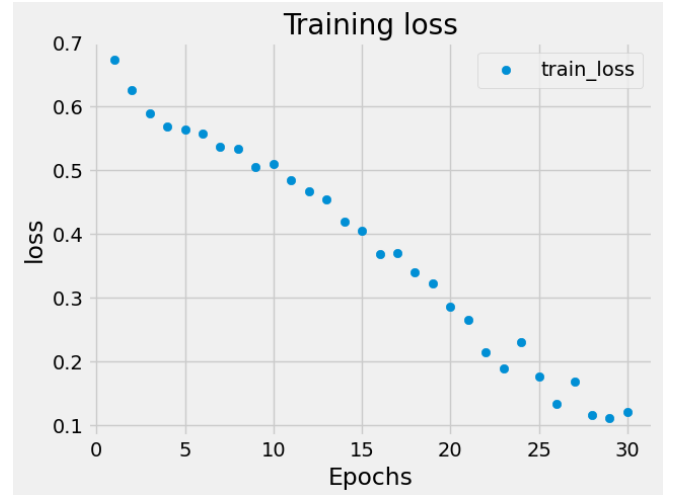


Figure 7 Training loss

Results comparison

For the water classify system, the Neural Network model reached a higher level of recall, precision and F1-score at (0.9) with (0.95) accuracy. As for SVM it is clearly that the precision where at (0.78), the recall and F1-score at (0.73), and the accuracy is (0.8).

The results indicate that the Neural Network model demonstrated better performance than the SVM model in forecasting water potability, exhibiting higher accuracy, precision, and recall scores. Overall, the Neural Network model proved to be more proficient in categorizing water samples as either potable or non-potable compared to the SVM model.

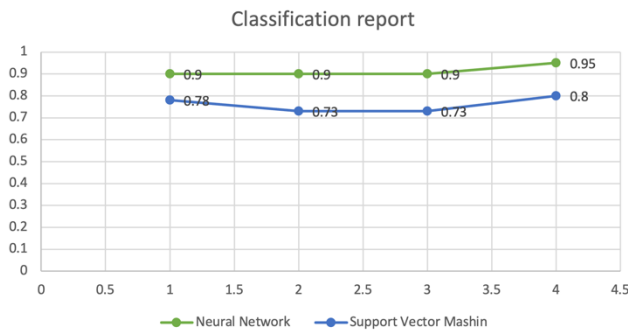


Figure 8 Water classify system

Confusion Matrix :

A confusion matrix is a table used to evaluate the predictive accuracy of a machine learning model by comparing its predicted and actual values. It consists of four essential values: TP, FP, TN, and FN. These values represent the counts of correct and incorrect predictions made by the model. The confusion matrix forms the foundation for calculating several performance metrics, such as accuracy, precision, recall, and F1 score.

	Predicted Positive	Predicted Negative
Actual Positive	233	29
Actual Negative	89	52

V. CONCLUSION

Our water quality prediction model holds promise for applications in the realm of water quality monitoring. In the future, it could be leveraged to develop an online tool that empowers users to monitor the quality of water in their local water bodies. This approach involves the collection of pertinent data using sensors to conduct the prediction. The subsequent phase of refining the water quality prediction model would entail obtaining the necessary stream data for accurate predictions. This would be achieved through a dynamic model update process. We obtain the results by our model in terms of water quality evaluation. The proposed

model was able to reach satisfying results in terms of accuracy, f1 score, and recall.

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